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RESEARCH-ARTICLE

New Frontiers of Technology. Leveraging Advanced Digital Tools for Effective Cultural Heritage Communication Engagement and Preservation

ALESSANDRO BASSO

CATERINA PALESTINI, University of G. d'Annunzio Chieti and Pescara, Chieti, CH, Italy

MAURIZIO PERTICARINI, University of Padua, Padua, PD, Italy

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New Frontiers of Technology. Leveraging Advanced Digital Tools for Effective Cultural Heritage Communication Engagement and Preservation

Alessandro Basso
Saad Scuola di Ateneo di Architettura
e Design
Camerino, Italy
alessandro.basso@unicam.it

Caterina Palestini
Università degli Studi G. d'Annunzio
Dipartimento di Architettura
Pescara, Italy
caterina.palestini@unich.it

Maurizio Perticarini
Unipd - Università degli Studi di
Padova
Padova, Italy
maurizio.perticarini@unipd.it



Figure 1: Image generated using generative artificial intelligence (Krea AI) based on the prompt: Piranesian Architecture of a monumental building under construction, in a classical painterly style, set in a 17th-century southern Italian city. (authors' elaboration)

Abstract

The article provides an overview of the potential of advanced digital technologies in the field of cultural heritage representation and dissemination. The adoption of innovative tools such as interactive XR environments supported by Artificial Intelligence dynamics, educational Serious Games, and digital archives promises to overcome

traditional barriers in cultural communication. Research documents how these technologies can facilitate more effective and inclusive cultural transmission, transforming the way cultural heritage is perceived and interacted with by the public, contributing to its preservation and enhancement for future generations. As evidence of such techno-methodological advances, the expeditious survey of a real space using Nerf reconstructive algorithms exploiting progressive video data for AI training on Postshot Beta platform is proposed. The experiment offers insight into the ease of use and speed of application of such low-cost systems compared with the quality of congruent and profitable results.



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CCS Concepts

• **Applied computing** → Arts and humanities; Architecture (buildings); Computer-aided design.

Keywords

Generative AI, Digital Reconfiguration, Digital Twin, Innovation, XR Environments

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1 Introduction

The investigation presented in this essay focuses on innovative approaches to digitizing and conveying tangible and intangible Cultural Heritage, with the aim of making its dissemination more accessible. In particular, it aims to structure plans and methods useful for its strategic dissemination, contributing to the usability of Architectural Design and, more generally, to the safeguarding of cultural legacy for future generations. Aiming to overcome, through the use of technological apparatus, the critical issues related to the interpretation and understanding of graphic apparatus by a non-specialised public, in relation to stylistic features, canons of expression and methods of representation that are often misunderstood or misinterpreted, the third and current fourth digital revolution has greatly contributed to the development of technologies for virtualization of environments, now involving Machine Learning and Artificial Intelligence dynamics, supporting various fields, from medicine to education, and then spreading and being widely used in the field of museum and archival dissemination [1], in the survey of cultural heritage and in the representation of the existing and non-existing through gamification dynamics, as demonstrated by the progressive recent increase in Serious games with a cultural theme, progressively approaching an increasingly less mechanized approach and more humanized outcomes, especially in relation to the composition and creation of explorable models that involve human skills in their conception/construction, reflecting an artistic-aesthetic curation that approaches the "digital humanities" and that seem to expand far beyond the disciplines that regulate human-computer interaction or machine-machine communication but refer, in effect, to a new system of Human-Human communication through innovative tools with a strong technological impact. It is thus a new kind of cultural transmission inherent in the very concepts of Information Mapping (acquiring input from the real or unreal) and Information Sharing (making such information usable fluidly to a variety of targets). The essay, therefore, offers an overview of the digital creation of spaces and artifacts by exploiting "cloning" features from the real and their communication through a Representation that can be integrated furthermore with current and advanced Virtual Reality (VR), Augmented Reality (AR) and Extended Reality (XR) devices, pursuing the goal of strategically disseminating the most diverse types of visual configurations derived from a survey of a real artifact, possibly from digital

modelling from Architectural Archive drawings, or drawn from descriptive and interpretable narratives, overcoming the difficulties of structuring adequate dissemination to a non-specialised audience. We highlight, for example, how interactive Archives in VR can mitigate these challenges, emphasising usability and enhancing the interpretation of virtualized artifacts (Digital Twin) or 3D reconfigurations of designs that were never built or no longer exist. Current technology, empowered in part by new AI approaches, thus facilitates the construction of new dimensions in which the concept of "culture" as a synthesis of both material heritage (landscape, cities, monuments and buildings) and intangible heritage (beliefs, traditions, knowledge, literatures and customs) is densified, while also simplifying its access and understanding thanks to new real-time Rendering systems that grant an intuitive and multi-experiential exploration that progressively improves in step with Computer Vision, in relation to the very evolution of digital technoculture.

2 Graphic-Digital Developments and New Media: the four Revolutions of Representation for the Valorisation of Cultural Heritage

Over the past fifty years, the field of architecture and design has experienced a series of radical transformations, mainly driven by graphic-digital developments. These transformations have led to a substantial change in modes, methodologies and operational processes in the creation of graphic designs, as well as a significant change in representation itself, triggering perceptions of real revolutions. The first two digital revolutions occurred in the last two decades of the 20th century. During this period, there was a transition from analogue representational modes-such as manual drawing, the use of colour on paper, and traditional drawing based on the rules of descriptive geometry-to digital modes of representation. The introduction of computer-aided design (CAD) drawing and three-dimensional rendering marked a turning point, enabling unprecedented precision and versatility in architectural design and visualization. From the mid-early 2000s to the present, two further digital revolutions have developed, with the fourth still in progress. These developments have involved the adoption of three-dimensional environments through more direct and immersive approaches, unconstrained by pre-structural direction. This has significantly reduced the distance between the user and the representation, allowing

for more immediate and engaging interaction. The third digital revolution saw the introduction of virtual reality (VR) devices and real-time rendering engines. These tools have greatly improved the ability to create photorealistic images and movies and to navigate three-dimensional environments without having to wait for the long rendering times typical of precomputed rendering. In particular, the use of HMD (Head Mounted Display) VR viewers has enabled users to fully immerse themselves in digital worlds, masking the visual perception of the surrounding physical world. However, despite the support of large multinational corporations, this technology has not yet achieved the hoped-for mass deployment. Technical limitations-such as the inconvenience of visors for prolonged sessions and the persistent problem of "motion sickness"

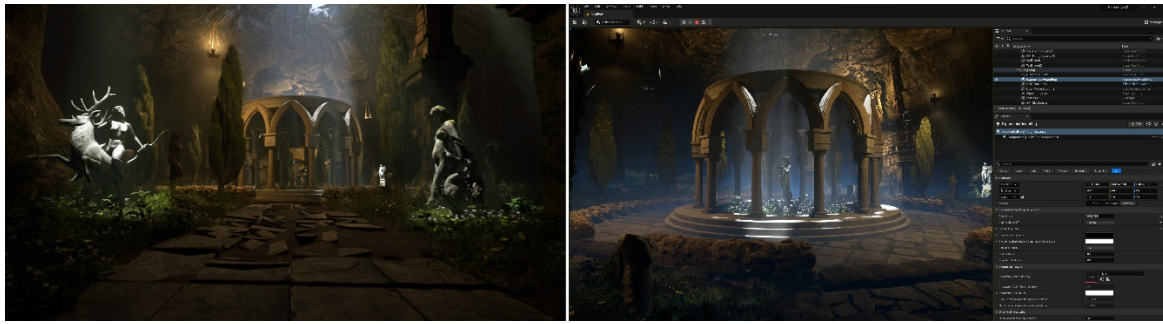


Figure 2: Images from the interactive academic project and thesis by Fabio Fazzini of SAAD, University of Camerino, on the implications of virtual reality for cultural dissemination related to the lyrics of Cecco D'Ascoli.

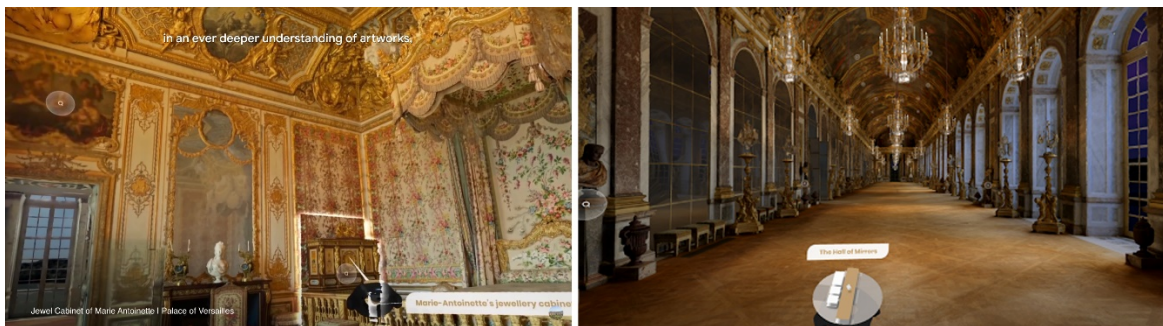


Figure 3: Google Art and Culture project Versailles VR: Images related to the 2019 multimedia cultural project that aimed to fully digitize the Palace of Versailles through a serious game with exploratory and educational dynamics in Virtual Reality, using Oculus and HTC Vive headsets.

and high production costs, as demonstrated by Apple's recent visor, are still significant obstacles. The fourth and most recent digital revolution, currently underway, is characterized by the massive use of artificial intelligence (AI) and machine learning in graphics production. The introduction of these technologies has profoundly changed various areas of representation, such as survey and image generation. AI offers substantial support in methodologies and new graphical approaches, automating complex processes and improving the accuracy and quality of representations. However, the use of AI algorithms raises many ethical questions, especially regarding the involvement of huge amounts of external data required for AI training. This includes graphic material, photographic and pictorial images with defined copyrights that, once used for training, lose their original meaning and make it difficult to trace the original source. The integration of AI into various sectors represents an unstoppable revolution that requires the implementation of complex regulations. These regulations must facilitate the ethical and proper use of AI while promoting experimentation and a thorough understanding of the medium. Only through appropriate regulation and a understanding of AI's potential and limitations will it be possible to fully exploit its innovative capabilities, while ensuring that authors' rights are respected, and the original meaning of works used for training is preserved. In the context of Cultural Heritage, with reference to the digital revolutions mentioned above,

the application of the dynamics of Information Mapping and Information Sharing plays a central role: these two approaches aim at conveying detailed information related to Heritage directly within any kind of immersive Digital Representation, allowing the user to learn by exploring virtual environments and interacting with various cultural elements, with the goal of not forcing choices (as may happen in a cinematic sequence in which a direction is preponderant) while remaining as little pedagogical as possible and thus leaving an almost total freedom of inquiry. Today, the technological tools in our possession thus allow us to interface with photorealistic scenarios that are almost indistinguishable from real ones, greatly expanding the illusion of interacting with an existing artifact, while at the same time giving us the possibility of enriching such digital environments with those Information Mapping contents that were fundamental to the very construction of the scenario and that can be employed directly within the same three-dimensional space, offering the possibility of accessing, through direct or indirect query systems that can be activated near certain elements (Hotspots), further insights without interrupting the continuity of the exploratory flow. The principles of Information Sharing, on the other hand, are linked to a subsequent phase of disclosure of the content previously inserted actively or passively by the programmer architect (or rather, by the Level Design) within the virtual environment, which thanks to the new VR/XR Devices (Meta Quest 3 and Apple Vision Pro) is now more usable for that user who wishes to entertain with

it a close-to-reality, fluid and natural approach, taking advantage of ultra-high resolution lenses, three-dimensional audio, haptic vibrations and other performing features. The trend of recent times, and this applies to both serious games for cultural promotion, Meta-verse dynamics and virtual museums 2.0, turns out to be to develop an intuitive and minimal interface while maintaining a good level of photorealistic graphic perception. In fact, according to current research, getting closer to the perception of reality can improve content acquisition and memorization [2]. This is due to a better and more intuitive adaptability to learn elements, and generally explore the alternative 3D dimension. This principle applies to both the use of XR optical viewers and a less intrusive approach through "first-person" desktop mode exploration. Information Mapping and Information Sharing thus demonstrate efficacy both in the case of digital scans from real contexts, acquiring data from places existing at a given time, and in the case of three-dimensional Reconfiguration based on manual modeling, personal texturing, and analysis of original drawings, dictated by graphical input, a narrative, or even a subjective fantastical interpretation. The inclusion of significant elements makes it possible to immediately identify the historical period represented or the style of a particular architect, defining a visual graphic style that can make the digitization project more attractive. The relationship between the Cultural Property and its digital "Transfiguration" is thus closely influenced by a number of determining factors. These could include, for example, the intrinsic nature of the asset itself, its scale factor, which is closely related to how it is perceived by the target audience, and its actual purpose of use, i.e., why it is intended to transpose the asset into a dimension that can be perceived exclusively through technological apparatuses. The issue of accessibility, moreover, is crucial in determining to whom this digital transposition will be addressed and how it will be usable by different audience segments. Let us try to define an executive workflow that can encompass the principles of Information Mapping and Information Sharing implemented in the construction of a digital project.

3 New technological implications of Computer Vision for the digitalisation of Architecture

The advancement of hardware and software technologies, empowered by the improvement of graphics cards and processors, together with the spread of Artificial Intelligence systems, has greatly expanded the dissemination and accessibility of tools for capturing real space. This progress has established the basis for a precise methodological procedure in the creation and massification of digital cultural media. Before recording scalable material and geometric information from the real, however, it is necessary to go through a preliminary stage of documentation on the object of study, which includes a historical and visual analysis using analogue drawings, sketches, CAD drawings, returns from life, and other similar analogue techniques. Then, following a structured indirect survey plan, it is possible to proceed with the 'actual three-dimensional acquisition, through the different methodologies available today, such as laser scanner, Lidar sensors, terrestrial photomodelling or through UAV drones, or through the new Nerf/Gaussian Splatting methodology, in which machine learning and artificial intelligence facilitate and speed up several complex operations [3]. The next

phase is the processing of the acquisitions, a critical step that must take into account various factors such as the target audience, the possible level of optimization, the target platform, and how the data itself will be disseminated [4].

Recently, there is an emerging solution that allows direct navigation of digitized data without requiring further steps of polygonal simplification or optimization: thanks to the use of increasingly high-performance PCs, it is possible to read such information even in its embryonic stage of point cloud or through direct exploration in the form of a very dense polygonal model, thanks to algorithms such as Nanite from Unreal Engine 5 (Epic games) or thanks to the open source multiresolution format adopted by some web platforms such as CNR's Aton. New acquisition technologies make it possible to extrapolate precise geometric and colorimetric data regarding the Heritage of architectures, allowing us to interface with photorealistic scenarios, almost indistinguishable from reality, significantly improving the perception of being in contact with the existing artifact. This also offers the possibility of enriching such digital environments with Information Mapping content, which can be queried directly within the three-dimensional exploration space, providing access to additional information without interrupting the exploratory flow, approaching an H-BIM approach. On the other hand, direct acquisition related to information mapping, where data are generated at the same time as the scans and not entered later by the programmer, is mainly useful to engineers and architects for Heritage conservation operations, which internationally today is in a particularly critical phase due to risks related to climate change and wars, which can destroy it irreparably. In a wartime scenario, the only advantage lies in the ability to predict in advance where conflicts will move, allowing volunteers and cultural associations to protect artworks and generate Digital Twins of them. In this context, our work plays the crucial role of digitizing the artifact to generate a spatial model as faithful as possible to the original, leaving a trace of itself and preserving its geometric form through detailed data acquisition, which is crucial for possible future reconstructions. A resource that makes these purposes explicit can help underscore the importance of these new technologies for enhancement by using them as an acquisition and dissemination tool, for example, on interactive virtual archives and museum platforms. In a phase of Information Sharing, that is, the dissemination and, in essence, the fruition of content inserted within the graphic programming structure, these are more attractive to the user who has a more fluid and natural interaction with them, more easily accessing notions on multiple overlapping educational levels thanks to immersive devices that are progressively more convenient and therefore capable of increasing the time of exploratory sessions. The quality of the experience is also becoming increasingly engaging, and the content itself benefits from the possibility of structuring itself into larger and more photorealistic virtual environments thanks to XR viewers, i.e., capable of bringing together virtual reality, augmented reality and mixed reality compatible on online platforms in a single device. We are thus witnessing an interesting phenomenon in which the same technology, initially developed to support non-interactive graphics (cinema, videomapping or precomputed animations), evolves by transcending its initial purposes, involving a wide range of domains [5]. Such involvement ranges from the medical sphere, using VR to treat phobias or educational purposes

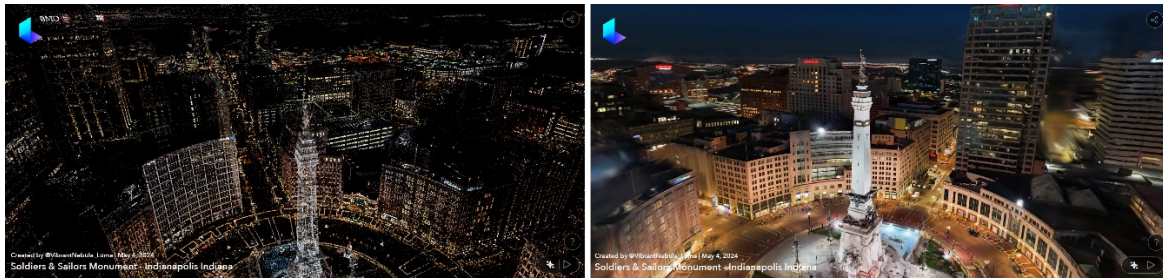


Figure 4: Digitization of an Indianapolis neighbourhood on the Luma AI platform using new real-space digitization systems with NeRF and Gaussian Splatting AI algorithms, capable of generating a voxel cloud based on image or video data. The model is exportable in various formats. (author: @VibrantNebula)



Figure 5: Images from the interactive academic project and thesis by Fabio Fazzini of SAAD, University of Camerino, on the implications of virtual reality for cultural dissemination related to the lyrics of Cecco D'Ascoli. The project is titled "Ink & Lyrics: Unravelling the Poets of Love."

in medical training, to engineering for simulations of structural calculations, to architecture and the generation of applications for heritage enhancement and management of three-dimensional digital archives [6]. In both contexts, both Information Mapping and Information Sharing can thus pursue the common goal of transmitting and communicating content to non-specialised users.

4 Communication Progress: Serious Game as an Alternative for Synthesis and Reinterpretation of Digital Cultural Heritage

In the creative process, the main goal is to convey the feeling and memory of a specific place, without necessarily replicating all its elements exactly. This does not imply the generation of an

exact digital copy (Digital Twin), but rather the restitution of the feeling, with appropriate simplifications, of being in a specific place. This particular quality can be recognized in the new interactive media, specifically in video games. Technological advancement has enabled video game studios to develop increasingly complex, engaging and interesting products, significantly expanding their audience compared to the past [7]. This advancement has not only elevated the video game medium to the same level as film, television and other communication systems, but has also contributed to its massification while becoming a new art form. Video games are no longer confined to purely recreational purposes but are used in a wide range of contexts. A significant example of this evolution is



Figure 6: Leartes Studios Historical reconstructions. The Art team consists of very experienced Environment Artists, able to create any types of environments in Game Engines such as Unreal Engine and Unity. (image from <https://leartesstudios.com/>)

gamification, a system that, although it can be addictive, is being used in areas as diverse as work.

Amazon, for example, uses gamification to make its workers' performance system competitive. Many companies adopt this approach to improve employee services through scoring mechanisms. This system, while having negative aspects, stimulates productivity. Indeed, video games are based on structures that incentivize competition and score accumulation, but if designed correctly, they can offer significant cognitive benefits. Recent studies have shown that exploring virtual environments can promote problem solving and stimulate memory, highlighting the versatility of video games in various fields of application. Not only do they encourage competition, but they can also improve reasoning and memorisation skills. In recent years, video gaming has evolved significantly from simple two-dimensional point-and-click games that were limited to challenge and point accumulation. Today, modern video games offer advanced graphical experiences and virtual spatiality that are seamlessly integrated with game dynamics, making them attractive even to a general audience. Famous series such as "Final Fantasy" and "Assassin's Creed" demonstrate how the environmental element is crucial to the success of a video game, being closely linked to the narrative and gameplay. Virtual reality (VR) technology, popularized during the third digital revolution, has further improved the approach of the latest generation of video games, making the user an active protagonist of the virtual dimensions [8]. Through three-dimensional viewers, realistic and interactive virtual environments can be explored, enhancing game dynamics and stimulating attraction to digital content, as well as cognitive skills related to information acquisition and memory. Over the past 15 years, commercial video games have reached very high levels in

terms of quality and performance. In parallel, the "serious game," a type of video game with educational purposes, which uses the same dynamics as commercial video games for strictly educational purposes, has become widespread. The serious game is one of the most widely used tools in academia, science and museums for cultural dissemination. Some virtual museums and digital architecture archives also use dynamics specific to serious games, introducing virtual reality applications and photorealistic three-dimensional graphics to make the experience more immersive. As a new digital media of communication, it empirically demonstrates the crucial importance of the graphic aspect of a virtualized environment, which can be perceived as real, and the freedom of movement within it. These elements facilitate cognitive connections and logical reasoning, optimizing mnemonic abilities and stimulating the curiosity of exploration. This thus implies significant added value when the serious game is applied to education concerning respect for and preservation of cultural patrimony, strengthening emotional connections and engaging the sentimental aspects that can preserve its existence. These factors make the serious game an ideal and instrumental resource for promoting and enhancing both tangible and intangible heritage. Therefore, the construction of a serious game cannot disregard the quality of the representation and the impact that at the level of scenario and narrative this has on the user. Increasingly complex virtual environments and advanced modelling tools, accompanied by digital model optimization systems, improve the construction of these games, making them in their production cheaper, faster and less laborious. These educational tools thus become progressively more attractive and widespread, expanding their use in diverse disciplinary fields. Since 2005, numerous experiments have been conducted and prototypes of serious games

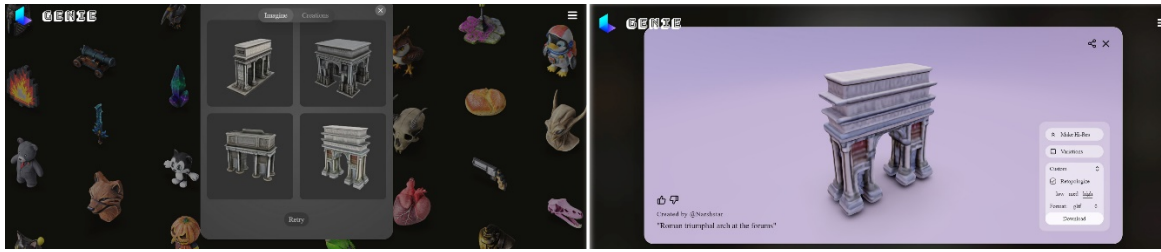


Figure 7: Experimental 3D modelling using the Luma AI Genie application, capable of generating 3D objects in various topologies and resolutions, exportable in multiple formats. The guidance is provided via text prompts LLM model. The generated models can vary in character and scale, ranging from characters to furniture, architecture, and decorative elements. (authors' elaboration)

developed for cultural patrimony enhancement, such as Rome Reborn and Ancient Pompeii. These projects ushered in a new form of involvement in the areas of communication, visual expression of information, mechanisms of collaboration between player avatar and virtual avatar, interactivity and playful entertainment. The “Rome Reborn” project represents one of the most comprehensive projects of digitization and interactive reconfiguration of historical urban scenarios. The main objectives of the project included the creation of an HD version of Rome in 320 AD, a low-resolution model for integration with Google Earth (<http://earth.google.com/roma/>), and a collaborative mode of the same model in high resolution for virtual reality (VR) applications intended primarily for university education. To study the effectiveness of the “Rome Reborn” project as a serious game for learning, exploring, re-enacting and researching the cultural and architectural aspects of Ancient Rome, the application evolved into “Roma Nova”. This was commercialized and used in various components in the development of other VR and AR applications. The project aimed to test the suitability of this technology to support archaeological exploration, with a focus on historical, cultural and social aspects of Roman life, such as political, religious and artistic expressions [1]. A latest updated version was released in early 2024, and it is still the most comprehensive in content and the most graphically impressive.

5 Integration with new virtualisation systems based on the use of machine learning

In recent years, there has been a remarkable spread of machine learning tools and, more generally, artificial intelligence-based technologies to improve services in various application areas. This includes not only new natural language systems (LLM - Large Language Model), which allow computers to generate text almost indistinguishable from that produced by humans, such as ChatGPT, but also tools in the field of graphic and artistic creation, such as Midjourney or OpenAI’s DALL-E 3, which allow even non-expert users to generate highly realistic photographic images or drawings via simple written prompts. Currently, there are two main systems that involve artificial intelligence in the graphics context. The first uses generative intelligence, such as the GAN (Generative Adversarial Networks) algorithms, which basically create images out of thin air based on an internal dual AI system working in competitive collaboration, while the second focuses on improving the acquisition of

spatial data from the real world, comparable to image-based digital sensing systems. In the creative process of building virtual spaces, these methodologies are frequently integrated both in the concept art phase, to define the atmosphere and visual style of scenarios, and in the concrete creation of 3D assets and objects. In particular, for the creation of 3D objects from descriptive prompts without the support of manual modelling, the Luma AI community recently released an innovative system called “Genie”, which applies the principles of two-dimensional image generation to three-dimensional objects (<https://lumalabs.ai/genie?view=create>) through a textual command prompt. In creating a 3d space, although these objects are not yet perfected, they can be used to create furniture or decorative elements, enriching the scene and giving it more credibility and character. Currently, creating an entire scene using only LLM systems (and thus textual instructions and natural language instead of design scripts) is problematic because of the control difficulties arising from full automation and dependence on models loaded into the artificial intelligence system. An effective solution is to integrate these new techniques with traditional modelling techniques, such as Nurbs or polygonal modelling. Alternatively, it is more effective to create environments by taking advantage of new AI acquisition systems from the real world, comparable as mentioned before to image-based self-modelling or even range-based scanning. These systems, such as Nerf virtualisation and Gaussian Splatting, make it possible to create realistic, though not directly editable, assets. Software such as Luma AI and Polycam, which are still under development, use proprietary external servers to generate exportable three-dimensional objects in specific formats, such as Luma Field with the Nerf algorithm and “Interactive Scene” with the Gaussian Splatting algorithm, either directly in Unreal Engine 5 or by conversion to already mapped and textured .obj/.GLTF formats. Other software, such as Instant Nerf or Postshot, make local use of GPU and CPU computing power to generate point clouds and voxel clouds in various common formats, such as .Ply. However, the main criticality of these systems lies in the inability to modify native elements and the difficulty of fitting them correctly within a complex scenario. In general, the use of such Assets within interactive scenarios is advantageous for representing distant elements that are not subject to direct interactions [9]. Approaches involving artificial intelligence applications in surveying and measurement operations represent a significant advance over traditional image-based or range-based digital sensing

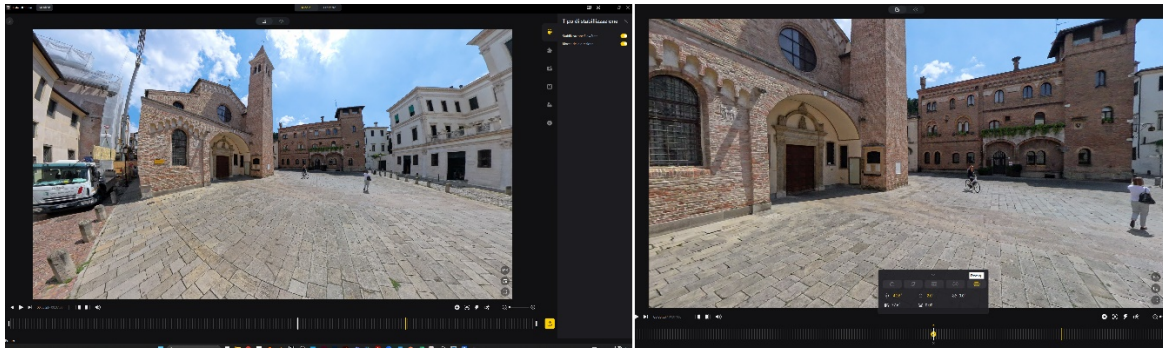


Figure 8: Digital AI Survey. Acquisition and Processing phase from spherical videos conducted using InstaStudio, proprietary software from the camera manufacturer. This software was crucial for making video cuts and eliminating image distortions caused by the two fisheye lenses using the "dewarp" command. This process enabled the creation of multiple sequences from orthogonal viewpoints, essential for minimizing errors in the data input for AI model training. (authors' elaboration)

methods. However, challenges remain, especially about millimetre error management. Currently, 3D photogrammetry continues to be the most accurate method for precise measurement operations. However, tools offered by AI machine learning already provide significant support, simplifying data acquisition operations and generating versatile representations that capture the atmosphere and structure of real-world elements. These methodologies provide remarkable photorealistic rendering, especially when applied in three-dimensional contexts, such as virtual platforms for video games (Unreal Engine and Unity), supported by specific plugins for implementing such content. Despite doubts about the potential implication of artificial intelligences in generating graphical-visual simulations indistinguishable from reality, there is a gradual and inexorable spread of this technology, which is finding more and more applications in Computer Vision. This diffusion takes place both in algorithmic-generative forms, as in the case of Midjourney for generative 2D art or Luma 3D, and as a re-targeting tool in the insertion of 3D models into Augmented Reality (AR) dynamics, exploiting a kind of evolution of the SLAM algorithm. Such an approach speeds up and simplifies the construction and management phases of interactive three-dimensional environments, offering a wider audience the opportunity to contribute to the structuring of complex virtual worlds, compatible with the 3D digital archives system [10].

6 Case study: simplification and innovative AI automation for the digital survey of the Piazzetta di San Nicolò in Padua

Computer vision, in conjunction with technological advancements in graphics and enhanced hardware capabilities, has enabled a substantial evolution both methodologically and in the ease of execution. This progress is particularly evident in the speed of algorithm application, pertaining to the digital reconstruction of real spaces as well as the reconfigured reconstruction of imaginary spaces. These reconstructions are also applicable to interpreting spaces designed on paper, transforming two-dimensional representations into exploratory three-dimensional visualizations. This transformation significantly enhances the relationship between human-subjective

interpretations of spaces and environments and the direct use of digital tools. These tools are empowered and optimized by the application of artificial intelligence and real-time rendering software. Furthermore, the integration of computer vision techniques has revolutionized the architectural and design industries by providing more accurate and efficient methods for spatial analysis and visualization. The ability to convert 2D plans into 3D models allows for a more immersive and interactive experience, aiding designers and architects in better understanding and communicating their projects. This advancement not only improves the accuracy of the designs but also facilitates a more intuitive and user-friendly interaction with the spatial data. Additionally, the synergy between AI and real-time rendering technologies has paved the way for innovative applications in various fields such as virtual reality, augmented reality, and digital twin environments. These applications are instrumental in creating highly detailed and dynamic representations of physical spaces, thereby enhancing decision-making processes and improving overall project outcomes. The continuous development in this area promises further enhancements in the fidelity and realism of digital reconstructions, pushing the boundaries of what can be achieved in digital space visualization and interaction. As evidence of such techno-methodological advances, the expeditious survey of a real space using Nerf reconstructive algorithms exploiting progressive video data for AI training on Jawset's Postshot Beta platform is proposed. The experiment offers insight into the ease of use and speed of application of such low-cost systems compared with the quality of congruent and profitable results. The presented case study is related to the reconfiguration of Piazzetta San Nicolò in Padua, consisting of historically significant buildings such as Palazzo Meschini and Palazzo Brunelli-Bonetti, as well as the Romanesque church of San Nicolò dating back to 1088. The church underwent several modifications starting from the 14th century, culminating in the restoration carried out between 1966 and 1971. The space is articulated, especially near the church where there is the bell tower and the entrance framed by an arch and set back from the façade. To perform the survey, the Insta 360 x4 camera was used in 8k video mode at 30 fps. With the aid of a one-meter pole, the video started from the entrance of the square

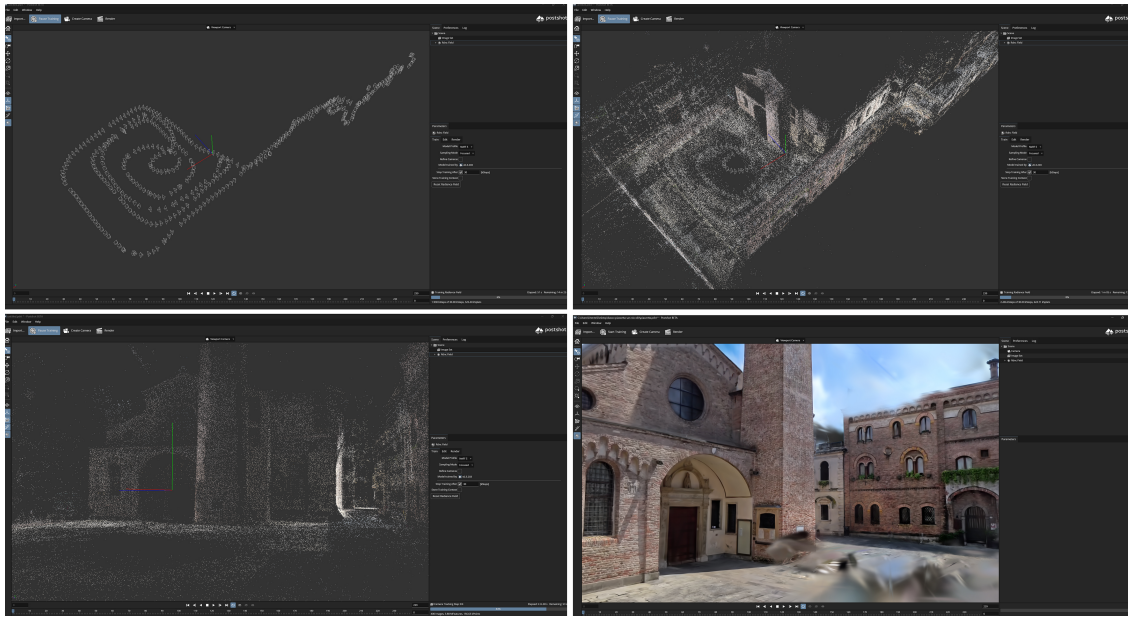


Figure 9: Digital AI Survey. Acquisition and Processing phase from spherical videos conducted using InstaStudio, proprietary software from the camera manufacturer. This software was crucial for making video cuts and eliminating image distortions caused by the two fisheye lenses using the "dewarp" command. This process enabled the creation of multiple sequences from orthogonal viewpoints, essential for minimizing errors in the data input for AI model training. (authors' elaboration)

on the Via Dante side — both to have a precise starting point to return to and to also capture the Loggia Bombardieri, which serves as the entrance to the square — proceeding with a slow and uniform pace and keeping the camera about two and a half meters from the ground level. The executed path followed a circular and concentric trajectory that developed from the outside (near the building façades) to the inside (centre of the square), then back to the starting point. Once the video was taken, the acquisition and processing phase took place using InstaStudio, the proprietary software of the camera's manufacturer. It was essential for making video cuts, eliminating image distortions due to the two fisheye lenses with the "dewarp" command. The cuts were exported from the native 360 video by locking the direction and identifying the four cardinal points and a zenithal direction from bottom to top to also capture the overhangs and ceilings (entrance of the church and Loggia Bombardieri). The next phase involved loading the five videos obtained from the processing into Jawset's Postshot software in beta version 0.3.203, available for free. On the approximately 63,000 frames derived from the five videos, each 7 minutes long, the software allows the establishment of the actual number of frames to consider for NeRF training. A value of 1000 was set. Through the software interface, it is possible to determine the quality of the NeRF and the processing duration in samples (30k as the default value). The results demonstrate the digital model's fidelity to the real-world artifact in terms of scale and proportion, exhibiting a low error percentage given the size of the analysed space. The efficiency and simplicity of execution further underscore its accuracy. Additionally, the model shows remarkable versatility in managing output data, which can be utilized across various external editing

platforms. These platforms can enhance, clean, and edit voxel or Gaussian clouds, with online applications such as Super-Splat serving as prime examples. In essence, the case study aimed to highlight the ease of managing digital space generation operations in a digital twin mode compared to the previous generation's low-cost methods [11]. The findings confirm that digital twin technology offers a significant improvement in the accuracy and efficiency of spatial analysis. This advancement is particularly important for industries requiring precise digital reconstructions, enabling better decision-making and more refined project outcomes. The digital model's ability to integrate seamlessly with advanced editing tools further enhances its utility, demonstrating the potential for widespread application and adaptation in various sectors.

7 Conclusions

In summary, technology and Computer Vision thus represent tools that currently serve cultural dissemination, creating new spaces capable of containing vast amounts of data. These tools can transcend the traditional educational sphere, proving useful in situations such as post-disaster or post-conflict reconstructions, as well as in the restoration of historical artifacts. Virtual space takes the form of a museum 2.0, an explorable and infinitely extended realm in which the essences of culture converge, ranging from design to architecture, archaeology to other cultural fields. At the state of the art, Computer Vision-supported technology has made remarkable progress, significantly influencing the field of architectural representation and the multiple disciplines of surveying. In addition to the introduction of precise modelling tools and advanced management of three-dimensional models, rendering techniques, both

precomputed and real-time, have contributed substantially to the creation of highly photorealistic representations of architectural artifacts. Recently, new algorithms have enabled the massive handling of large numbers of polygons and textures, achieving levels of detail unthinkable just a few years ago. Moreover, with the introduction of dynamic digital lighting, the simulation of light and shadow, taking advantage of accurate management of reflection and refraction, without resorting to gimmicks such as light baking (which blended light effects onto textures but at the same time made scenery extremely static) is now able to reproduce the real behaviour of photons in a detailed and physical way. The use of new shaders, known as BPR (Physically Based Rendering) materials, offers the ability to realistically perceive a wide range of materials, providing precise control over their physical characteristics. This has reduced the need for artifacts to render more detailed objects in virtual space, as current hardware and software allow them to be handled at full resolution. A notable contributor to this rapid acceleration in technical development is the creation of new graphics GPUs, such as Nvidia's RTX series, and the increase, both in number and quality, of the computational units, the cores, in the processors relative to the hardware itself. This comprehensive premise on digital evolution in support of contemporary representation lays the foundation for new frontiers and innovative performance in support of the archiving and subsequent dissemination of Heritage, both tangible and intangible [12]. This technological advance allows a broad spectrum of scholars to employ such tools to structure an interactive archive, in online or offline multimedia mode, while maintaining a high standard of quality, compatible with a plurality of media. This approach facilitates, for example, the use of open-source software such as Aton, developed by researchers at the CNR, or of highly professional solutions employed in the field of Applied Game and commercial video game development for entertainment, such as Twinmotion or Unreal Engine 5 from Epic Games. These methodological and technological advancements are revolutionizing not only the way projects for Heritage valorisation are designed but also the very execution methodology. They are progressively moving towards solutions that are so simple and automated that they allow everyone, potentially leveraging crowdsourcing techniques, to input useful datasets into multimedia multiplatform systems. Additionally, they enable anyone, even without modelling knowledge, to produce digital models that can be accessed in real-time and shared objectives on the web. At this point, new regulations are urgently needed to structure such methodologies based on computer intelligence, as they could, on a large scale, lead to inaccuracies and errors if left unchecked.

References

- [1] Basso Alessandro. 2017. La realtà virtuale come metodologia di sviluppo di nuove figure professionali: gestione e condivisione del Cultural Heritage in ambito museale, riabilitazione cognitiva e progettazione architettonica. In: *UID2017 Napoli Territori e Frontiere della Rappresentazione*, Gangemi Editore, Roma, 1617-1624.
- [2] Coco Alberto. 2020. La grande industria del videogioco per scuole e musei: l'esperienza di Ubisoft con Assassin's Creed. In: Pescarin Sofia, Ed., *Videogames, Ricerca, Patrimonio Culturale*, Franco Angeli, Milano, 43-47.
- [3] Kerbl Bernhard, Kopanas Georgios, Leimkuhler Thomas, and Drettakis George. 2023. 3D gaussian splatting for real-time radiance field rendering. *ACM Trans. Graph.* 42, 4, 1-14.
- [4] Fanini Bruno, and Cinque Luigi. 2020. Encoding immersive sessions for online, interactive VR analytics. *Virtual Reality*, 24, 3, 423-438.
- [5] Montagna Lorenzo. 2018. *Realtà Virtuale e Realtà Aumentata, nuovi media per nuovi scenari di business*. Edizioni Hoepli, Milano.
- [6] Luigini Alessandro, Brusaporci Stefano, Kofler Werner, Basso Alessandro, Maiezza Paola, Tata Alfredo, and Rossi Maria Luisa. 2023. The Vaults of the Church of St. Michael the Archangel in Brixen between Geometry, History and Missed Space. *Nexus Network Journal*, 1-10.
- [7] Allison John. 2008. History educators and the challenge of immersive pasts: a critical review of virtual reality tools and history pedagogy. *Learning, Media and Technology*, 33, 4, 343-352.
- [8] Annetta Leonard A., Mangrum Jennifer, Holmes Shawn Y. and Collazo Kimberly. 2009. Bridging reality to virtual reality: Investigating gender effect and student engagement on learning through video game play in an elementary school classroom. *International Journal of Science Education*, 31, 8, 1091-1113.
- [9] Palestini Caterina, Basso Alessandro, and Perticarini Maurizio. 2022. Machine Learning as an Alternative to 3D Photomodeling Employed In Architectural Survey And Automatic Design Modelling. *International Archives of Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII, 191-197. DOI: <https://doi.org/10.5194/isprs-archives-XLVIII-2-W1-2022-191-2022>.
- [10] Suneetha V, Salini Suresh., Niharika Sinha, Sabyasachi Prusty., and Syed Jamal. 2020. Enhancement in the World of Artificial Intelligence. *International Research Journal on Advanced Science Hub*, 2, Special Issue ICARD 2020, 8, 276-280.
- [11] Palestini Caterina, Meschini Alessandra, Perticarini Maurizio, and Basso Alessandro. 2024. Neural Networks as an Alternative to Photogrammetry. Using Instant NeRF and Volumetric Rendering. In: Giordano Andrea, Russo Marcello, and Spallone Raffaella, Eds., *Beyond Digital Representation. Advanced Experiences in AR and AI for Cultural Heritage and Innovative Design*, Springer, Cham, Switzerland, 471-482.
- [12] McCullough Malcolm. 1998. *Abstracting Craft: The Practiced Digital Hand*. MIT Press, Cambridge.
- [13] Basso Alessandro. 2020. *Virtual Environments for new media*. Aracne Ed., Roma.
- [14] Basso Alessandro. 2020. New interpretation tools and metamorphosis of the image, how the self-synthesizing of visual elements influences the aesthetic evolution. In: Cicalò Emanuele, Ed., *Advances in Intelligent Systems and Computing 1140; Proceedings of the 2nd International and Interdisciplinary Conference on Image and Imagination IMG 2019*, Springer Nature, Basilea, 923-935.
- [15] Basso Alessandro. 2019. Serious game and 3d modelling for the immersive experience's development in digital tale. In: Parrinello Sandro, Ed., *Digital & Documentation Databases and Models for the enhancement of Heritage*, Pavia University Press, Pavia, 68-79.
- [16] Basso Alessandro, Condorelli Francesca, Giordano Andrea, Morena Sara, and Perticarini Maurizio. 2024. Evolution Of Rendering Based On Radiance Fields. The Palermo Case Study For A Comparison Between Nerf And Gaussian Splatting. *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-2/W4-2024, 57-64. DOI: <https://doi.org/10.5194/isprs-archives-XLVIII-2-W4-2024-57-2024>.
- [17] Checa David, and Bustillo Andres. 2020. A review of immersive virtual reality serious games to enhance learning and training. *Multimedia Tools and Applications*, 79, 9, 5501-5527.
- [18] Guo Shiyu, Sapatnekar Sachin, and Gu Jie. 2024. Physical-Based Ray-Tracing Rendering Processor for Photorealistic Augmented Reality with Inverse Rendering and Background Clustering for Mobile Devices. In *2024 IEEE International Solid-State Circuits Conference (ISSCC)*, 44-46. DOI: <https://doi.org/10.1109/ISSCC49657.2024.10454394>.
- [19] Luigini Alessandro, Fanini Bruno, Basso Alessandro, and Basso Demis. 2020. Heritage education through serious games. A web-based proposal for primary schools to cope with distance learning. *VITRUVIO, International Journal of Architectural Technology and Sustainability*, 5, 2, 1-12. DOI: <https://doi.org/10.4995/vitruvio-ijats.2020.14665>.
- [20] Laurent Alexandre. 2018. *La guerra delle intelligenze: Intelligenza artificiale contro intelligenza umana*. Edizioni EDT, Torino.
- [21] Park Keunhong, Sinha Utkarsh, Barron Jonathan T., Bouaziz Sofien, Goldman Dan B., Seitz Steven M. and Martin-Brualla Ricardo. 2021. Nerfies: Deformable Neural Radiance Fields. In *Proceedings of 2021 IEEE/CVF International Conference on Computer Vision (ICCV)*, IEEE, Piscataway, NJ, 5845-5854.
- [22] Remondino Fabio, Karami Ali, Yan Ziyang, Mazzacca Gabriele, Rigon Simone, and Qin Rongjun. 2023. A Critical Analysis of NeRF-Based 3D Reconstruction. *Remote Sensing*, 15, 14, 3585. DOI: <https://doi.org/10.3390/rs15143585>.
- [23] Schönberger Johannes L., and Frahm Jan-Michael. 2016. Structure-from-motion revisited. In *IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2016*, IEEE, 4104-4113.